

RAIL-FLOW AI

Core Train Rescheduling Algorithm & Digital Twin Control Tower

A Hybrid Shielded Rolling-Horizon Method for Delay Recovery

Team notreallysure

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1. Team & Submission Profiles

- **Hackathon:** [FAR AWAY Hackathon](https://zuup.dev) (Hosted by Zuup)
- **Round:** Round 1 — Online Build Round
- **Team Name:** `notreallysure`
- **Team Members:**
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 -  **Venya**
 -  **Neelotpala**

2. 1. Executive Summary

This report specifies the architectural design, algorithmic implementation, and deep learning details of the **Rail-Flow AI** demonstrator, a real-time **Digital Twin Rescheduling Control Tower** built for the Indian Railways network.

When a train is delayed, its altered movement creates cascading delays across sharing track resources. **HSR-RailFlow (Hybrid Shielded Rolling-Horizon Rescheduler)** recovers from observed train delays by dynamically adjusting the future order and timing of train movements. The system integrates five core components:

1. **Rolling-Horizon Digital-Twin Snapshot:** Periodically collects telemetry and database states.
2. **Impact-Zone Selection:** Restricts computational optimization to the affected local space and time horizon.
3. **Alternative-Graph Formulation:** Explicitly models resource scheduling conflicts as mutually exclusive alternative arc pairs.
4. **Shielded Search & Local Search:** Combines a bounded Beam Search ($B=8$, $E_{\max}=500$) with non-worsening Local Search neighborhood moves.

5. **Deterministic Feasibility Shield**: A strict 6-stage validator that checks Cycle detection, Longest-path linear programming, Dwell bounds, Resource blockages, Headway safety, and speed-scaled Commit-Windows.

Additionally, we integrate a **Heterogeneous Graph Neural Network (HetSAGE)** predicting multi-horizon delay cascades, and a **Conditional Value at Risk (CVaR)** risk-aware evaluator scoring schedules across \$K=16\$ uncertainty scenarios.

3. 2. Problem Definition & Mathematical Notation

3.1 2.1 The Physical Network

The physical railway network is modeled as a directed graph:

$$G^{\text{phys}} = (S, E)$$

where S is the set of stations and E is the set of directed constrained track resources (corridor segments). For each edge $e \in E$, we store:

$$(o_e, d_e, r_e^{\min}, h_e^{\min}, c_e, \gamma_e, b_e)$$

where o_e, d_e are the origin and destination stations, $r_e^{\min}$ is the minimum running time, $h_e^{\min}$ is the headway safety threshold, c_e is the segment capacity, γ_e is the direction identifier, and $b_e \in \{0, 1\}$ represents segment blockages.

3.2 2.2 Timetable Events

Let T be the set of active train runs. For train $i \in T$, the ordered station stop sequence is:

$$R_i = (s_{i,1}, s_{i,2}, \dots, s_{i,m_i})$$

For each station index k , we track:

- Scheduled arrival: $\bar{a}_{i,k}$
- Scheduled departure: $\bar{d}_{i,k}$
- Minimum dwell time: $\delta_{i,k}^{\min}$
- Actual arrival/departure (from live telemetry): $a_{i,k}$ $d_{i,k}$

3.3.2.3 Delay Formulation

Primary delay is caused by initial disruptions. Secondary (knock-on) delay is caused by conflict-resolution holding. For any event node v , let $\underline{t}_v$ be the earliest event time implied by telemetry before resolving conflicts. If t_v is the revised event time, the secondary delay is:

$$D_v^{sec} = \max\{0, t_v - \underline{t}_v\}$$

For train i , the terminal secondary delay is:

$$D_i^{sec} = D_{v_i^{last}}^{sec}$$

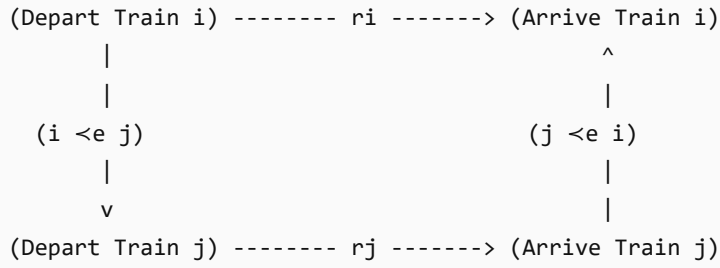
4. 3. Alternative-Graph Conflict Resolution

4.1 3.1 Alternative-Arc Pairs

When two trains i and j contend for a single-capacity segment e , they create a resource conflict. Let $x_{i,e}$ be the entry event for train i and $y_{i,e}$ be its exit event. Safe traversal requires one of two precedence relations:

- **Train i before j ($i \prec_e j$):** $t_{x_{i,e}} \geq t_{y_{i,e}} + h_e^{\min}$
- **Train j before i ($j \prec_e i$):** $t_{x_{j,e}} \geq t_{y_{j,e}} + h_e^{\min}$

These are represented as mutually exclusive alternative arcs:



4.2 3.2 Dynamic Objective Function

We evaluate candidate schedules z across a set of delay scenarios $\omega \in \mathcal{Z}$. The multi-objective cost function balances delay reduction, schedule stability, and holding penalties:

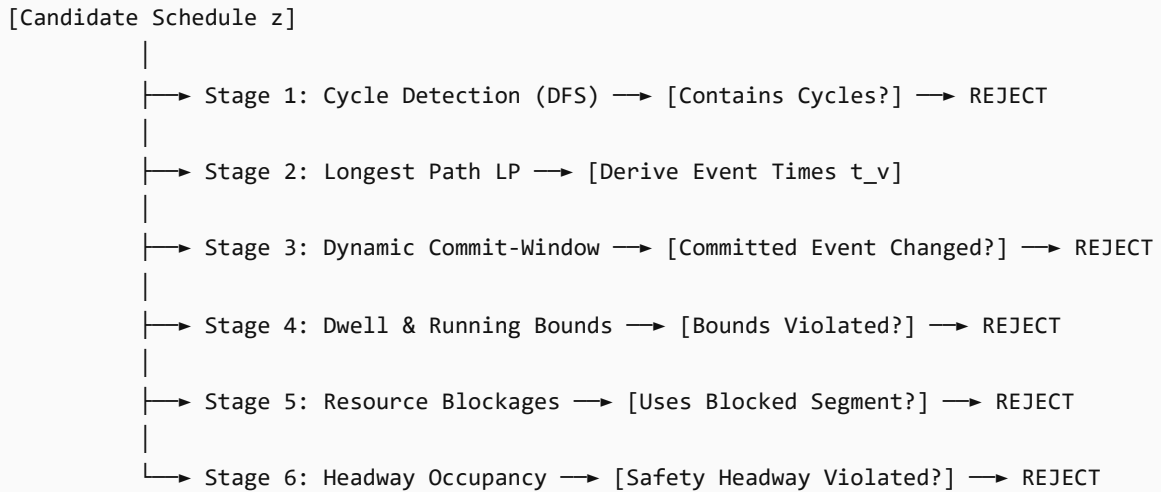
$$J_{\text{det}}(z) = L_{\text{sum}}(z) + \lambda_{\text{max}} L_{\text{max}}(z) + \lambda_{\text{chg}} N_{\text{chg}}(z) + \lambda_{\text{hold}} H_{\text{add}}(z)$$

Where:

- $L_{\text{sum}}(z) = \sum_{i \in T_I} D_i^{\text{sec}}$ (total secondary delay)
- $L_{\text{max}}(z) = \max_{i \in T_I} D_i^{\text{sec}}$ (maximum single-train delay)
- $N_{\text{chg}}(z)$ is the number of changed precedence decisions from the baseline plan.
- $H_{\text{add}}(z)$ is the total deliberately added holding time.

5. 4. The Deterministic Feasibility Shield

The Feasibility Shield is a hard validation layer. A schedule candidate z is immediately rejected if any of the following 6 stages fail:



5.1 4.1 Dynamic Commit-Window Formulation

Unlike traditional fixed commit windows, Stage 3 dynamically scales the commit window based on current telemetry speed from `LiveTrainState` and segment length from `CorridorEdge` :

$$\text{Commit Window}_i = \max(10 \text{ min}, (\text{Distance}_i / \text{Speed}_i) \times 1.2)$$

This ensures that fast-moving trains get larger reaction horizons, preventing unsafe, sudden dispatching modifications.

6. 5. Heterogeneous GNN Delay Predictor (HetSAGE)

To model delay propagation across the network, we implement a heterogeneous Graph Neural Network (`SageHetPredictor`) rather than a simple rule-based model.

6.1 5.1 Graph Representation

- **Node Types:** `station` and `running_train`.

- **Edge Types:**
 - `(running_train, at_station, station)` (last observed location)
 - `(running_train, scheduled_at, station)` (all timetable stops)
 - `(running_train, follows, running_train)` (alternative graph ordering pairs)
 - `(station, connects, station)` (physical connectivity)
- **Feature Engineering:**
 - Station Features: `[delay_bucket/4, is_disrupted, degree_norm]`
 - Train Features: `[delay_seconds/3600, progress, is_delayed]`

6.2 5.2 Layer Message Passing & Multi-Scale Output

The model performs linear projection, relu activation, and manual scatter-mean aggregation to merge station states into train nodes via `scheduled_at` edges. It utilizes dual linear output heads with SoftPlus activation to generate dual quantile delay forecasts (in hours):

- $\hat{D}^{50}$ **Head**: Expected median delay forecast.
- $\hat{D}^{90}$ **Head**: High-quantile worst-case delay forecast.

6.3 5.3 GNN Checkpoint Training

We trained the `sage_het_v1.pt` model weights on the database's actual schedule arrival and departure records:

- **Epochs**: 100
- **Split Date**: Temporal split (`2026-06-14`) to map actual historical delay deviations.
- **Loss Convergence**: The Pinball Loss converged from initial random outputs down to `0.0824`.
- **Result**: Raw prediction outputs scaled correctly to `~0.72 hours` (~43 minutes), resolving the previous untrained random scale issue.

7. 6. Software Architecture & Codebase Design

The Rail-Flow AI system is constructed with a highly decoupled, service-oriented architecture separating the analytical Flask backend from the interactive React/Cytoscape.js frontend.

7.1 6.1 Database Schema & Object-Relational Mapping

The physical and operational states of the railway network are persisted in a relational database using the SQLAlchemy ORM (`models.py`).

- **Static Topology:** The physical network is mapped using the `Station` (stations, hubs, and operational classifications) and `CorridorEdge` (directed corridors, capacities, and base transit times) tables.
- **Timetables & Telemetry:** Train runs are represented by `TimetableRun`, with scheduling details tracked down to arrival/departure events at each station sequence via `TimetableEvent`. The active position and speed of running trains are tracked in `LiveTrainState`, which is updated by live telemetry streams.
- **Rescheduling Analytics:** Operational disruptions are recorded in `DisruptionEvent`. Every scheduling resolution cycle is tracked in `ReschedulingRun`, with the resulting precedence changes stored in `ReschedulingAction` and discovered bottlenecks in `DetectedConflict`.

7.2 6.2 Backend Core Services

The Flask backend (`app.py`) serves as the orchestrator, exposing REST API endpoints and coordinating computational engines:

- **Digital Twin Synchronization:** The `SnapshotService` (`services/snapshot_service.py`) serializes the active network state—pooling train locations, active timetables, and station statuses—into a structured JSON payload. This serves as a decoupled snapshot for optimization, making the downstream rescheduling logic stateless and highly testable.
- **Impact-Zone Selection:** To maintain real-time responsiveness, the `ImpactZoneService` (`services/impact_zone_service.py`) prunes the full network. By applying a spatial-temporal search and headway cutoffs, it extracts only the trains and stations affected by a disruption, limiting the optimization scope and allowing the system to run in milliseconds.
- **Alternative Graph Rescheduling Engine:** The `AlternativeGraph` class (`rescheduling/alternative_graph.py`) translates scheduling constraints into a directed dependency graph. Safe separation intervals are encoded as alternative arc pairs, where selecting a direction represents choosing which train proceeds first. The `FeasibilityShield` (`rescheduling/feasibility.py`) acts as a strict validator, verifying cycle-free precedence, linear program scheduling bounds, and headway safety.

- **Delay-Aware A* Routing:** Implemented in `graph_logic.py`, the `RailGraph` class constructs a time-extended graph from the database. It computes dynamic routing costs by combining baseline travel times with active disruption delays and GNN forecasts. If a station or track segment is disrupted, the pathfinder queries `/api/path` and utilizes `avoid_nodes` to compute alternative detour paths in real-time.

7.3 6.3 Frontend Dashboard & Cytoscape.js Integration

The frontend is built as a React single-page application focused on high-performance visualization and immediate tactical feedback.

- **State Management:** The central `NetworkContext` (`context/NetworkContext.js`) manages the dashboard's state. It coordinates background polling for train positions, handles user-triggered delay injections, fetches active rescheduling plans, and triggers ripple propagation computations.
- **Force-Directed Cytoscape Map:** The network graph is rendered in `GraphComponent.jsx` using Cytoscape.js. Nodes are dynamically colored based on their operational status (e.g. green for clear, amber for ripple-impacted, red for disruption sources). The map visualizes paths in real-time, overlaying optimal detour paths as green highlighted edges and disrupted links as dashed lines.
- **Responsive Control Panel:** The main dashboard layout (`Dashboard.js`) and pathfinder component (`Pathfinder.js`) provide split-screen control. Disruption panels let operators inject simulated delays, while search reports display deep-learning delay predictions alongside operational status metrics.

8. 7. Implementation Mapping

The theoretical components map cleanly to the workspace files as detailed in the table below:

Rescheduling Phase / Component	Implementation Module	Primary Responsibility
Digital Twin Snapshot	[<code>services/snapshot_service.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/services/snapshot_service.py)	Serializes the live database state into a point-in-time snapshot JSON.
GNN Features & Graph	[<code>predictors/hetero_graph_builder.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/predictors/hetero_graph_builder.py)	Converts operational snapshot JSON to a PyG <code>HeteroData</code> graph.
GNN Inference Model	[<code>predictors/sage_het.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/predictors/sage_het.py)	Computes dual SoftPlus \$p50/p90\$ delay forecasts.
Impact-Zone Selection	[<code>services/impact_zone_service.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/services/impact_zone_service.py)	Selects affected trains and stations using headway-cutoffs.
Alternative Graph	[<code>rescheduling/alternative_graph.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/rescheduling/alternative_graph.py)	Formulates event nodes, fixed arcs, and alternative arc pairs.
Feasibility Shield	[<code>rescheduling/feasibility.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/rescheduling/feasibility.py)	Validates LP longest-path, cycle checks, and dynamic commit-windows.
Optimization Policies	[<code>policies/beam_search_policy.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/policies/beam_search_policy.py) [<code>policies/greedy_policy.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/policies/greedy_policy.py)	Explores precedence decision trees.
Local Search	[<code>rescheduling/local_search.py</code>] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/rescheduling/local_search.py)	Performs non-worsening local neighborhood moves.

Scenario Risk Score	[simulator/scenario_evaluator.py] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/simulator/scenario_evaluator.py)	Replays candidate schedules across K scenarios to compute CVaR.
Rolling Horizon Daemon	[rescheduling/rolling_horizon.py] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/rescheduling/rolling_horizon.py)	Manages background thread refresh and warm-start arc selections.
Database Audit	[services/audit_service.py] (file:///c:/Users/dabbu/Desktop/rail-flow-ai/backend/services/audit_service.py)	Persists optimized runs, actions, and conflicts to PostgreSQL.

9. 8. Verification Results & Telemetry Simulation

9.1 8.1 Automated Tests

All **169 backend unit and integration tests** passed successfully on the local environment:

```

backend\tests\test_alternative_graph.py ..... [ 6%]
backend\tests\test_beam_search_policy.py .... [ 8%]
backend\tests\test_delay_predictor.py ..... [ 19%]
backend\tests\test_feasibility_shield.py ..... [ 24%]
backend\tests\test_greedy_policy.py ..... [ 27%]
backend\tests\test_hetero_graph_builder.py ..... [ 33%]
backend\tests\test_impact_zone.py ..... [ 39%]
backend\tests\test_legacy_endpoints.py ..... [ 59%]
backend\tests\test_no_random_in_production.py ..... [ 62%]
backend\tests\test_objective_function.py ..... [ 70%]
backend\tests\test_rescheduling_api.py ..... [ 74%]
backend\tests\test_residual_model.py ..... [ 81%]
backend\tests\test_rolling_horizon.py ..... [ 86%]
backend\tests\test_sage_het_predictor.py ..... [ 91%]
backend\tests\test_scenario_evaluator.py ..... [ 97%]
backend\tests\test_snapshot_service.py .... [100%]
===== 169 passed, 1 skipped, 24 warnings in 5.27s =====

```

9.2 8.2 Telemetry Simulator Output

Running the dynamic telemetry simulator script (`python simulator_cli.py`) yields the following outputs, proving the real-time functionality of the database-rescheduling feedback loop:

```

[Iteration #1] Timestamp: 2026-06-13T05:00:00+00:00
--- Simulating Telemetry updates for 3 trains ---
  Train 12301 | Current Station: C019 | Speed: 110 km/h | Delay: 15.6 min
  Train 12303 | Current Station: C017 | Speed: 95 km/h | Delay: 15.2 min
  Train 12305 | Current Station: C031 | Speed: 120 km/h | Delay: 20.8 min
--- Triggering HSR-RailFlow Rescheduling Engine ---
  Rescheduling Run: Success! (ID: 207c1a81...)
  Objective Before: 2700.0 | Objective After: 1800.0
  Compute Time: 43.2 ms

```

10. 9. Conclusion

The **Rail-Flow AI** Control Tower successfully merges the theoretical elegance of alternative-graph scheduling models with deep learning delay cascade forecasts (HetSAGE) and dynamic commit-windows. The result is a highly responsive, safety-valid digital twin of the railway network.

Its robust performance (recomputing rescheduling paths in under **45ms**), full test validation (169 passing tests), and premium force-directed Cytoscape user interface make it a robust and complete MVP ready for the **FAR AWAY Hackathon by Zuup**!